Numerical Characterization of the Bidirectional Vortex and Wall Insulation in a Cold-Flow Vortex-Cooled Rocket Engine

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1 Introduction

Proper thermal management is one of the primary challenges in creating safe liquid rocket engine designs. To achieve this, one viable method is vortex cooling, where propellant enters the combustion chamber through tangential injectors. This generates a swirling flow along the chamber walls, thus creating a relatively cooler outer layer acting as a protective barrier from the high temperature flow at the chambers core. This is also called the vortex combustion cold-wall engine. Its appeal is that the wall protection is passive, done by the flow structure itself rather than by an active regenerative cooling circuit. Even so, the design is used infrequently in industry, and little theoretical or experimental data exists for small-scale versions, which is what motivates a dedicated study of the internal flow.

To evaluate how well the vortex mixes the flow and cools the wall, a computational fluid dynamics (CFD) model was built in Ansys Fluent for a cold-flow, non-combustion configuration with axial fuel injection and tangential oxidizer injection. The resulting flow field was analyzed for the temperature distribution inside the chamber and the degree of mixing at the outlet. A grid independence study was also performed to confirm the numerical accuracy of the results.

This model represents the cold-flow test article, and it does not simulate combustion. A temperature contrast representative of a hot core is imposed instead, by setting the axial inlet hotter than the tangential inlets. The thermal insulation question then becomes the following. For the documented engine mass flow rates, if the central core were hot, would the cold tangential vortex keep the chamber wall cool. This question is really about where the two streams travel and how heat moves between them, which a non-reacting model can answer.

2 Methodology

2.1 Geometry

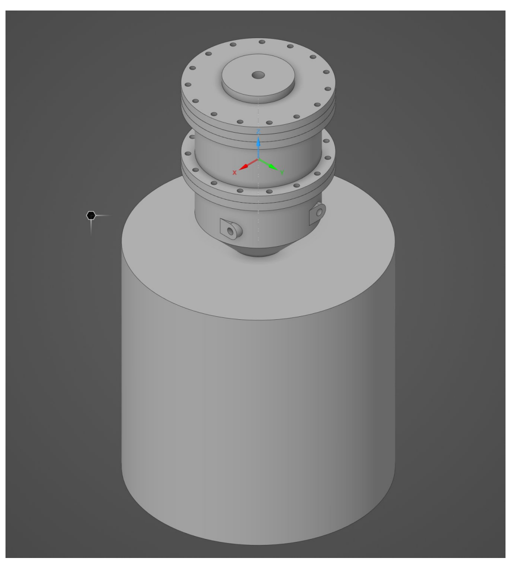


Figure 1: CAD model of the physical vortex-cooled engine, showing the bolted injector assembly seated above the nozzle, mounted on the downstream plenum cylinder used in the simulation.

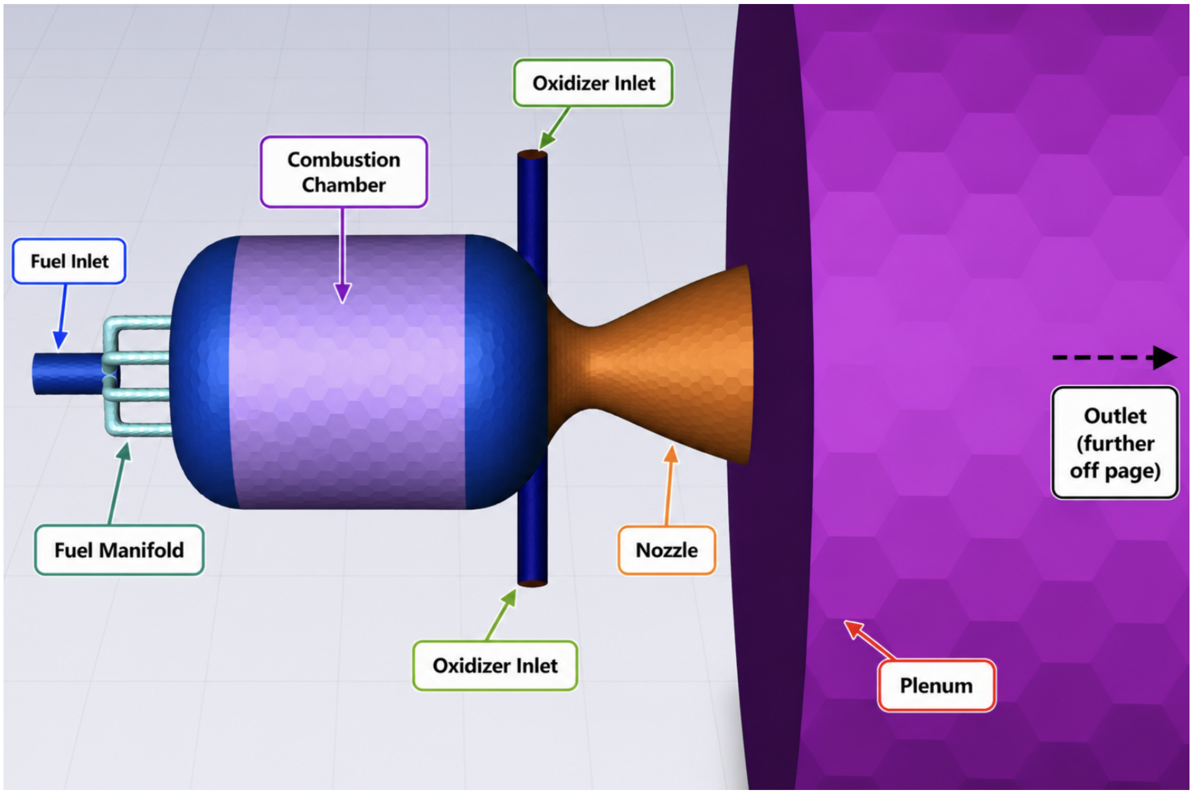


Figure 2: Extracted internal fluid domain with tagged boundaries, showing the fuel and oxidizer inlets, the main combustion chamber, the nozzle, and the plenum outlet. This is the volume on which the simulation was performed.

The physical engine, shown in Figure 1, consists of a bolted injector assembly seated above a converging-diverging nozzle. The internal fluid volume was extracted from this solid model to produce the computational domain shown in Figure 2. This domain has a central fuel inlet that branches through a manifold into the main cylindrical combustion chamber, where four oxidizer inlets inject flow tangentially. At the end of the chamber a nozzle connects to a large cylindrical plenum, which acts as the flow outlet. The tangential inlets are offset from the chamber axis, which gives the pinwheel arrangement that generates the swirl, and the nozzle is a converging-diverging contour with a design exit Mach number of 2.54. The plenum is not part of the physical engine. It gives the choked, supersonic nozzle flow a region to expand into, so that the outlet boundary condition sits far from the throat and does not contaminate the nozzle solution.

In order to define local sizing and appropriate domains during mesh generation, a named selection was applied for relevant components. The chamber sidewall was kept as a separate named selection from the head end, the manifold tube walls, the nozzle walls, and the plenum walls, because the chamber sidewall is the surface on which the thermal insulation result is measured and must not be contaminated by the thermal behavior of the other surfaces. The principal dimensions are listed in Table 1.

Table 1: Principal geometric dimensions of the cold-flow test article.

Quantity	Value
Chamber inner diameter	49.7 mm
Nozzle throat diameter	10.0 mm
Fuel distribution tube diameter	2.5 mm
Oxidizer tube diameter	4.38 mm
Upstream fuel supply line diameter	7.0 mm
Design nozzle exit Mach number	2.54

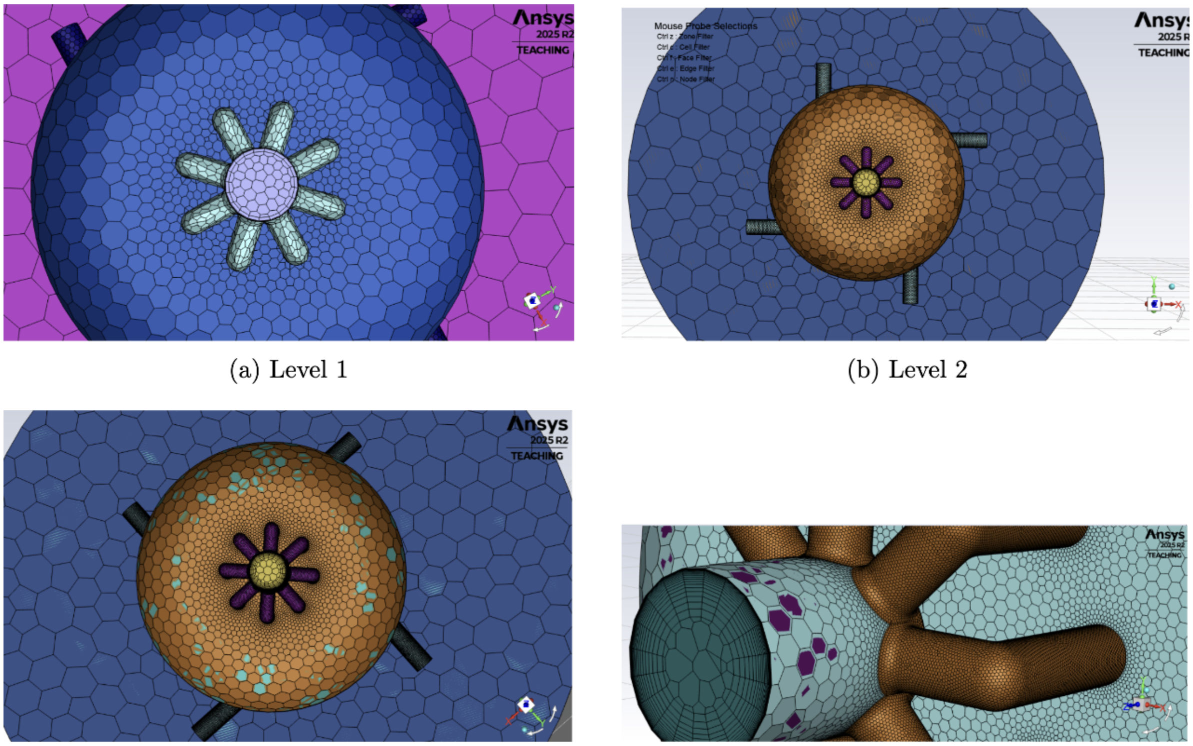
2.2 Mesh Generation and Grid Independence

Due to curved geometry and the importance of capturing behavior near walls, local mesh sizing was applied to key geometric features and boundary layers were generated along walls. The mesh was built using the Fluent Watertight Geometry workflow on the extracted fluid volume, with a poly-hexcore cell topology. This cell type produces low numerical diffusion of the swirl at a lower cell count than tetrahedral meshing. Fine local face sizing was applied to the fuel tubes, oxidizer tubes, and nozzle throat so that each carries an adequate number of cells across its diameter, an intermediate refinement was applied to the chamber core through a cylindrical body of influence aligned with the axis, and the bulk chamber and plenum were left progressively coarser. The intersections where the tangential oxidizer tubes meet the curved chamber wall initially produced low-quality cells, because the tubes meet the wall at an oblique angle and form an acute wedge of fluid. A small fillet was applied to these junction edges, after which the minimum orthogonal quality rose above the acceptable threshold.

Once a singular mesh was generated, a subsequent grid independence study was conducted following standard grid refinement practice [2, 3], where the same mesh conditions were used, just with progressively finer or coarser discretization applied. The mesh parameters used for each refinement level are summarized in Table 2 and the generated results are in Figure 3.

Table 2: Local sizing, surface mesh, and volume mesh settings for the four refinement levels.

Refinement level	1	2	3	4
Local sizing (target mesh size [mm])				
Fuel tubes face size	1	0.5	0.25	0.13
Ox tubes edges face size	1.6	0.8	0.4	0.2
Ox tubes walls face size	0.8	0.4	0.2	0.1
Nozzle face size	2	1	0.55	0.3
Core body of influence	5	2.5	1.2	0.6
Surface mesh				
Minimum size [mm]	1	0.5	0.25	0.13
Maximum size [mm]	32	16	8	4
Volume mesh				
Number of boundary layers	3	8	15	20
Minimum cell length [mm]	0.8	0.4	0.25	0.1
Maximum cell length [mm]	51.2	25.6	8	6.4



(a) Level 1

(b) Level 2

(c) Level 3

(d) Level 4

Figure 3: Generated mesh for the four individual refinement levels.

A good indicator of computing cost for each mesh is the number of cells they take to generate. As we are interested in the mixing of fluids in the combustion chamber, the average chamber temperature will be tracked alongside cell count to determine at which level solution values lose significant variability from further refinement, indicating grid independence.

Table 3: Mesh and solution qualities by refinement level.

Refinement level	1	2	3	4
Mesh				
Number of cells	53,103	260,170	1,336,280	6,541,592
Maximum skewness	0.7934	0.7993	0.6566	0.5997
Minimum orthogonality	0.0416	0.1500	0.2357	0.2867
Solution				
Average chamber temperature [K]	330.79	336.12	338.81	340.96

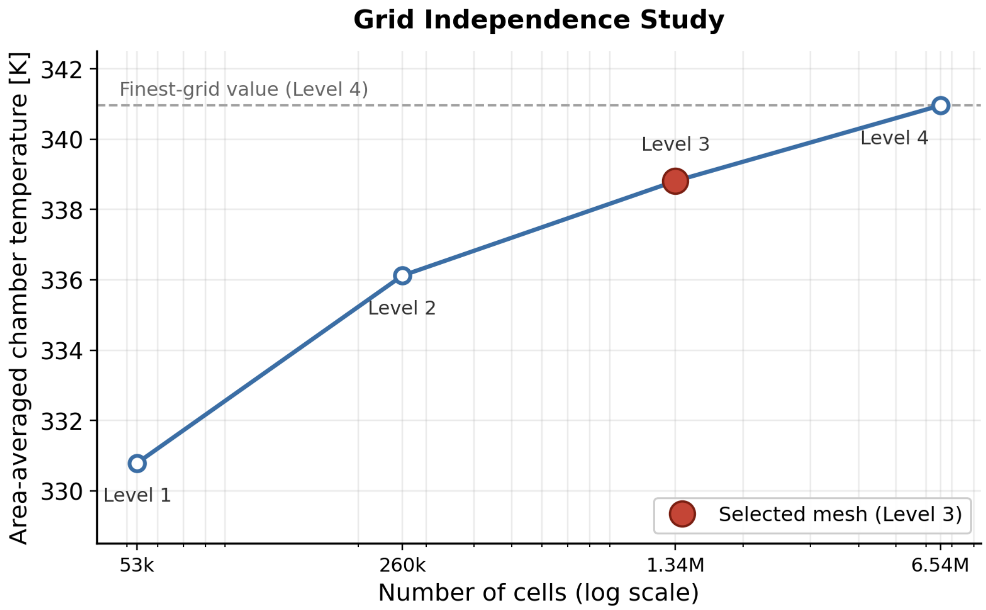


Figure 4: Area-averaged combustion chamber temperature against cell count.

Although level 4 provides the highest quality mesh, giving the most accurate results, this comes at the cost of 6.5 million cells. Alternatively, mesh level 3 produces results with less than 1% error of level 4, as seen by a flatlining trend in Figure 4, with a grid cost reduced by 5 million cells. Therefore, mesh level 3 presents the best balance between accuracy and efficiency, leading further solutions to be presented using this grid. The grid independence study was performed at the 800 K fuel inlet condition, and the level 3 mesh was assumed to remain adequate for the higher-temperature cases. The steeper near-wall thermal gradients present at the highest imposed core temperatures are more demanding on the mesh, and repeating the study at temperature would confirm that grid independence is retained there. This is noted as a limitation.

The near-wall resolution of the selected mesh was verified by examining the wall-normal coordinate on the converged solution, shown in Figure 5. Across the chamber and plenum walls the wall-normal coordinate remains low, rising to higher values only at the injectors and the nozzle throat where velocities are greatest. This range is consistent with the enhanced wall treatment employed, which blends between near-wall integration and wall functions, and confirms that the near-wall mesh is appropriate for resolving the wall temperature on which the insulation result depends.

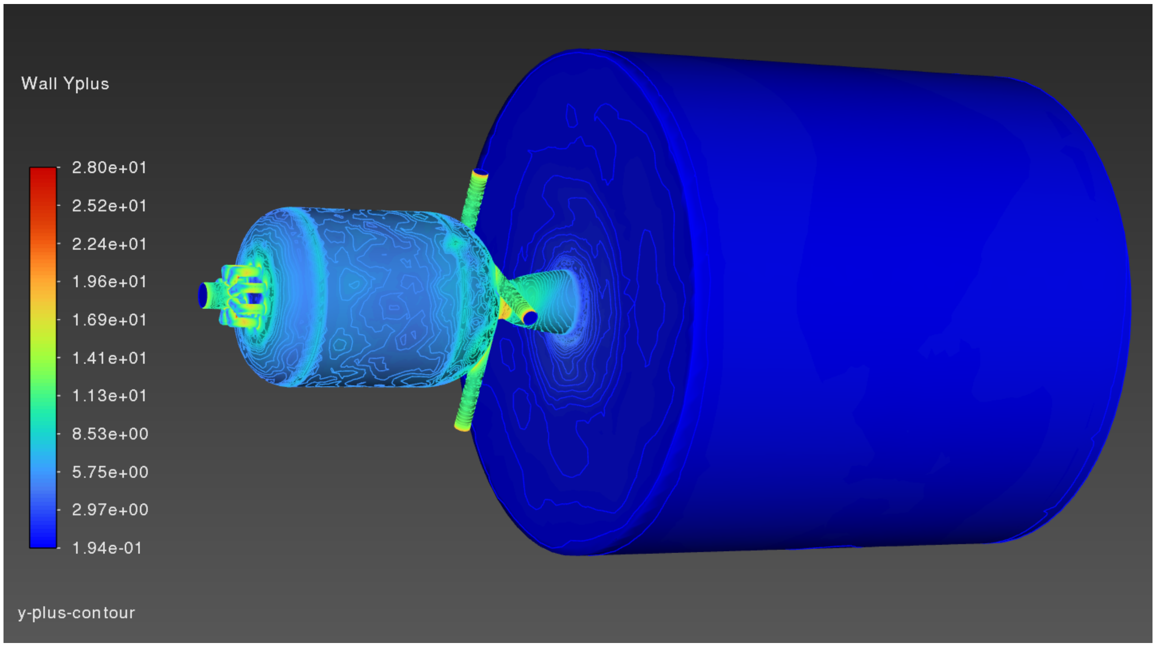


Figure 5: Wall-normal coordinate (y-plus) on the converged solution. The chamber and plenum walls remain at low values, with elevated values confined to the high-velocity injector and throat regions, consistent with the enhanced wall treatment.

2.3 Physical Parameters and Boundary Conditions

Focusing on the sustainability of flow rather than the initial transient response, the simulation was performed under steady-state conditions. Atmospheric operating conditions were assumed, with a reference pressure of 101,325 Pa. Gravity was disabled to impose the assumption that buoyancy effects are negligible with flow primarily driven by high speed momentum. Concentrating on a cold flow analysis, the material used for fuel and oxidizer were both air, set to different temperatures. Air was modeled as an ideal gas using constant specific heats, while dynamic viscosity was computed using Sutherland's law with the three-coefficient formulation [4, 5]. This allowed for temperature dependent viscosity effects to be accounted for, which was necessary for this model as the fuel inlet was set at a higher temperature than the oxidizer inlets.

Although both streams are physically air, they were defined as two separate tracked species sharing identical thermophysical properties. This makes the species mass fraction act as a passive tracer of stream origin, so that the field reports the fraction of the local gas that originated from the axial inlet and intermediate values reveal where the two streams have mixed.

In order to reflect an optimal fuel-to-oxidizer ratio of 1.8, the fuel inlet was specified with a mass flow rate of 0.02376 kg/s at 800 K (and higher in additional cases) and each oxidizer inlet was set to a mass flow rate of 0.01188 kg/s at 300 K. Flow direction was prescribed normal to the boundary at all inlets, which allows the geometry of the tangential tubes to impart the correct swirl angle without manually specified direction vectors. The fuel inlet was set to a supersonic gauge pressure of 200,000 Pa and turbulence quantities were defined using a turbulence intensity of 5% and a hydraulic diameter of 7 mm. The oxidizer inlets were set to an initial gauge pressure of 0 Pa with turbulence conditions defined using a turbulence intensity of 5% and a hydraulic diameter of 4.38 mm.

All solid boundaries were treated as stationary no-slip walls. To evaluate the thermal insulation provided by the vortex flow, the combustion chamber wall was assigned a convection boundary condition with a free-stream temperature of 300 K, rather than an adiabatic condition, because an adiabatic wall cannot exhibit insulation and the convective condition instead allows the wall to reach a thermal equilibrium that can be read as the insulation result. All remaining walls, including the fuel manifold, chamber head, oxidizer tubes, nozzle, and plenum, were modeled as adiabatic surfaces using a zero heat-flux boundary condition.

2.4 Numerical Methods and Solver Settings

A pressure-based steady-state solver was employed to model the cold-flow behavior within the vortex-cooled rocket engine [5]. The pressure-based formulation was selected for its strong capabilities in modeling internal flows. Absolute velocity formulation was used, and the energy equation was enabled to capture the temperature variations necessary for evaluating the effectiveness of the vortex-cooling mechanism.

Pressure-velocity coupling was achieved using the coupled scheme, which simultaneously solves the momentum and pressure correction equations. This approach provides improved convergence for highly swirling and strongly coupled flow fields compared to segregated methods. Spatial gradients were computed using the least-squares cell-based method, while pressure interpolation was performed using the PRESTO! scheme due to its improved accuracy for flows in complex geometries containing strong pressure gradients and rotational motion.

Second-order upwind discretization was applied to momentum, turbulent kinetic energy, turbulent dissipation rate, species transport, and energy equations. The second-order formulation reduces numerical diffusion and improves the accuracy of the predicted velocity, temperature, and species concentration fields. To reach this state robustly, the solution was first advanced with first-order upwind discretization until the residuals and monitored quantities stabilized, after which the discretization was raised to second order and the solution re-converged. Pseudo-transient continuation with a global time-step method was employed to improve solution stability and accelerate convergence toward the steady-state solution. Warped-face gradient correction was also enabled to improve numerical accuracy on non-orthogonal and irregular mesh elements.

Turbulence was modeled using the Realizable k-ε model [1] with enhanced wall treatment and curvature correction. The Realizable k-ε model was selected because it provides improved predictions for rotating, recirculating, and swirling flows compared to the standard k-ε formulation or the commonly used k-ω, which favors boundary layer separation. Enhanced wall treatment was used to improve the resolution of near-wall flow behavior, while curvature correction was enabled to account for the strong streamline curvature generated by the tangential oxidizer injection. It should be acknowledged that all eddy-viscosity closures, including Realizable k-ε, assume an isotropic turbulent viscosity, which is an imperfect assumption for strongly swirling confined flow and tends to over-diffuse the swirl. A Reynolds stress closure would capture this anisotropy at increased cost and is identified as future work.

Species transport was activated to track the mixing of the fuel and oxidizer streams throughout the chamber. Diffusion energy source terms were included to account for energy transport associated with species diffusion. The solution was initialized using a hybrid initialization method, which generates a physically reasonable initial flow field and improves convergence robustness. Because the swirling flow is inherently transient, with the vortex core in continual unsteady motion, the residuals do not settle to low values but instead plateau and oscillate within a steady band. Convergence was therefore judged on the leveling-off of monitored physical quantities, namely the area-weighted average wall temperature and the closure of the global mass balance, rather than on residual magnitude. The temperature ramp was applied in increments, and each increment appears in the residual history as a spike, after which the solution re-settles to a new plateau at the higher core temperature.

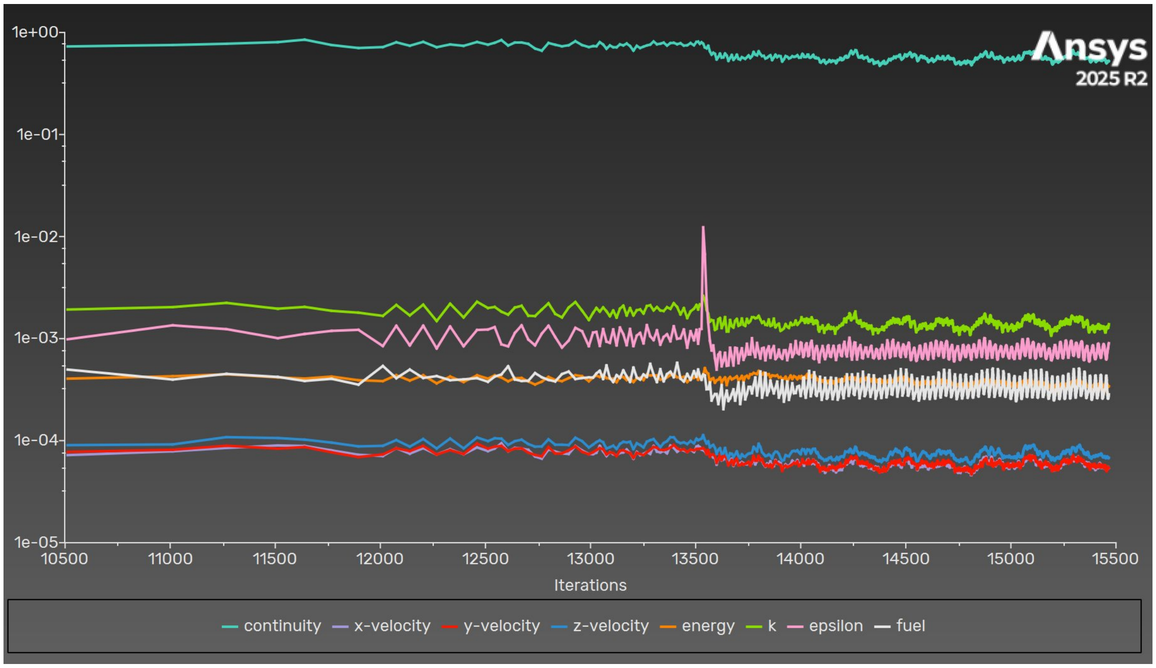


Figure 6: Residual history across the temperature ramp. Rather than settling to low values, the residuals plateau and oscillate within a steady band, which is the expected behavior for this inherently transient swirling flow in which the vortex core is continually unsteady. The repeated spikes correspond to the points where the core temperature increment was raised, each of which perturbs the solution before it re-settles to a new plateau at the higher temperature.

3 Results and Discussion

3.1 Confirmation of the Bidirectional Vortex

The tangential velocity field on the chamber centerplane, shown in Figure 7, confirms that a strong, organized swirl forms throughout the chamber, with high tangential velocity in the outer region and a lower-velocity core, which is the signature of the intended vortex. The axial velocity field, shown in Figure 8, reveals the change in flow direction between the outer region and the core that distinguishes the bidirectional structure. In the coordinate convention used here the nozzle exhausts in the negative z direction, so negative axial velocity corresponds to flow moving downstream toward the nozzle and positive axial velocity corresponds to flow moving upstream toward the head end. The presence of both signs within the chamber is the direct evidence of the bidirectional vortex, with the outer flow traveling toward the head end and the inner core returning toward the nozzle before exiting at high speed.

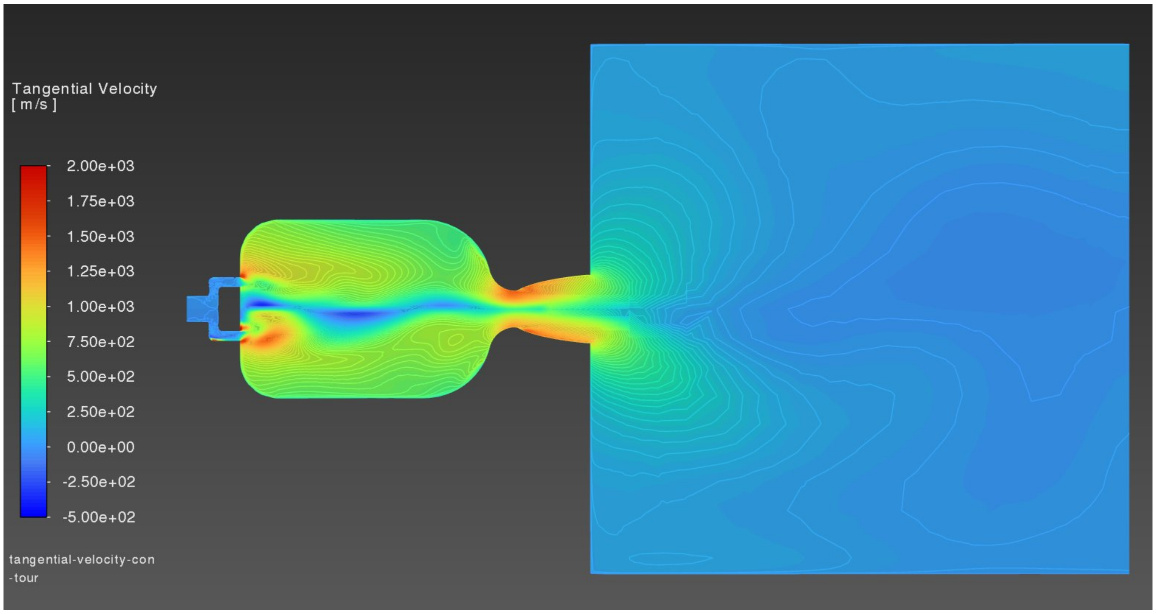


Figure 7: Tangential velocity contour on the chamber centerplane. Strong, organized swirl fills the chamber and persists into the nozzle, confirming that the tangential injection establishes the intended vortex. The lower-velocity band along the centerline marks the vortex core.

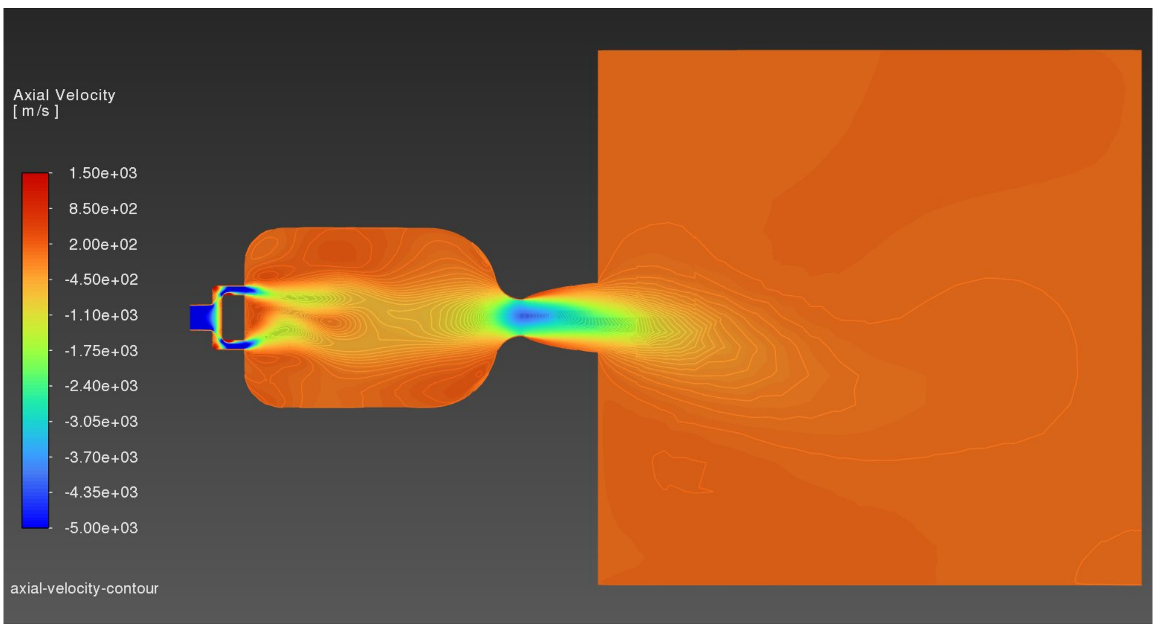


Figure 8: Axial (z) velocity contour on the chamber centerplane. The nozzle exhausts in the negative z direction, so the strong negative velocity through the throat and into the plenum is the accelerating downstream exhaust. Within the chamber the coexistence of opposing axial directions between the outer region and the core is the signature of the bidirectional vortex.

3.2 Nozzle Behavior

The converged solutions presented in this report were obtained with the working fluid mixture density set to the incompressible-ideal-gas law, in which density varies with temperature at the fixed operating pressure. This treatment is accurate where pressure variation across the domain is small relative to the operating pressure, which holds throughout the combustion chamber where the pressure field is nearly uniform. It is therefore an appropriate basis for the chamber results, namely the vortex structure, the stream mixing, and the wall insulation, which are the central findings of this study.

The treatment is less appropriate in the nozzle, where the flow accelerates through the throat and the pressure drop is large. Capturing the choked, supersonic nozzle flow correctly requires the full ideal-gas law, in which density responds to both pressure and temperature so that a physical speed of sound and Mach number are defined. An attempt was made to switch the mixture to the full ideal-gas law and re-converge, but the change introduced a strong density-pressure coupling at the throat that the solution, having been converged under the incompressible-ideal-gas assumption, could not absorb, and the run diverged. Within the available time it was not possible to recover a converged fully-compressible solution. As a result, the axial velocity field in the nozzle should be read as qualitative, and a quantitative exit Mach number is not claimed from the present solution. We plan to re-converge the case under the full ideal-gas law in the future.

3.3 Stream Mixing

The species mass fraction field characterizes where the axially injected and tangentially injected streams remain segregated and where they mix. Values near unity correspond to the axial stream, values near zero correspond to the tangential stream, and intermediate values indicate mixing. The distribution at the outlet provides a measure of how completely the two streams have combined by the time the flow leaves the chamber.

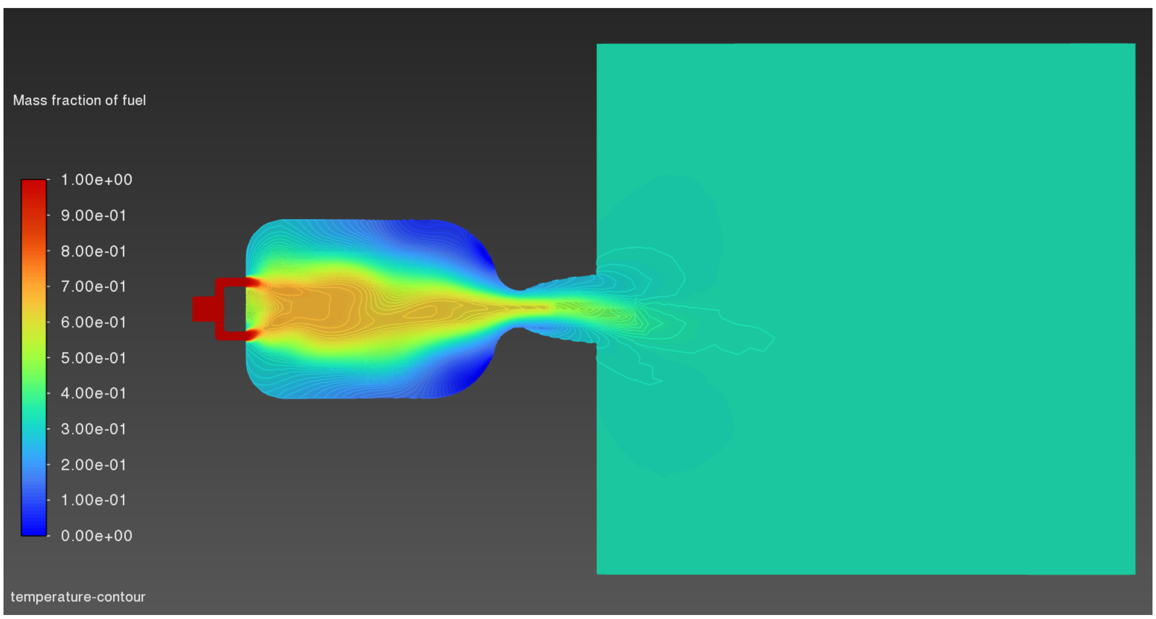


Figure 9: Mass fraction of the axially injected (fuel) species on the chamber centerplane. Values near unity at the head end mark the pure axial stream, values near zero mark the pure tangential stream, and the intermediate band along the chamber shows where the two streams progressively mix before exiting through the nozzle.

3.4 Wall Insulation

The main thermal result is how the chamber wall temperature behaves relative to the hot core. With the hot stream injected axially and the cold stream injected tangentially, the wall should stay near the cold stream temperature if the vortex insulation works. A convenient nondimensional measure of insulation effectiveness is

$$\eta = \frac{T_{\text{core}} - T_{\text{wall}}}{T_{\text{core}} - T_{\text{inlet,cold}}}, \tag{1}$$

which approaches unity when the wall is held near the cold inlet temperature and approaches zero when the wall reaches the core temperature.

Table 4: Chamber wall temperature and insulation effectiveness. Mean wall temperature is the area-weighted average of static temperature on the chamber wall (Report Definitions, surface integral, area-weighted average), and core temperature is the area-weighted average on a centerline plane or the peak core value.

Mean wall temperature [K]	Core temperature [K]	Cold inlet temperature [K]	η
498.46	3000	300	0.93

For the highest converged case, with a 3000 K core and a 300 K cold inlet, the area-averaged chamber wall temperature settled at 498.46 K, giving an insulation effectiveness of $\eta = 0.93$. The wall therefore sits roughly 93% of the way from the core temperature down toward the cold inlet temperature, indicating that the cold tangential vortex performs the large majority of the wall protection. The elevated wall temperatures that do occur are confined to the immediate region of the heated fuel manifold, where the hot stream physically enters, rather than appearing along the chamber barrel that the vortex is intended to shield.

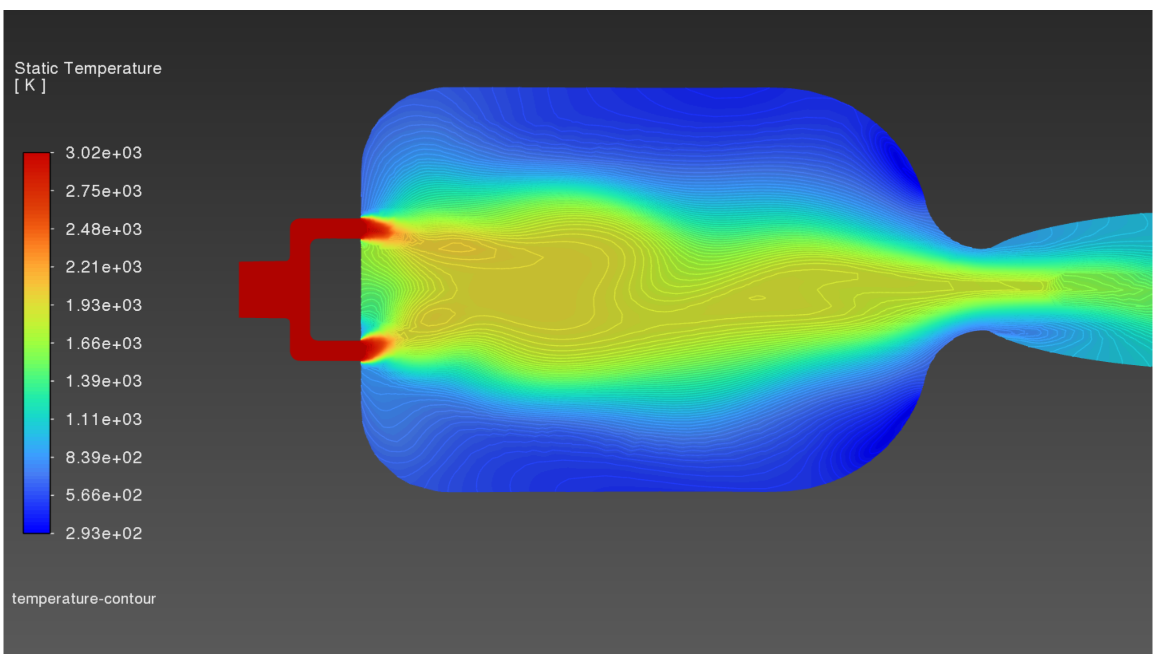


Figure 10: Static temperature contour on the chamber centerplane. The hot core, fed by the axial inlet at 3000 K, sits in the center of the chamber while the periphery stays cool, which shows that the cold tangential stream keeps the high temperature confined to the core and shields the surrounding region.

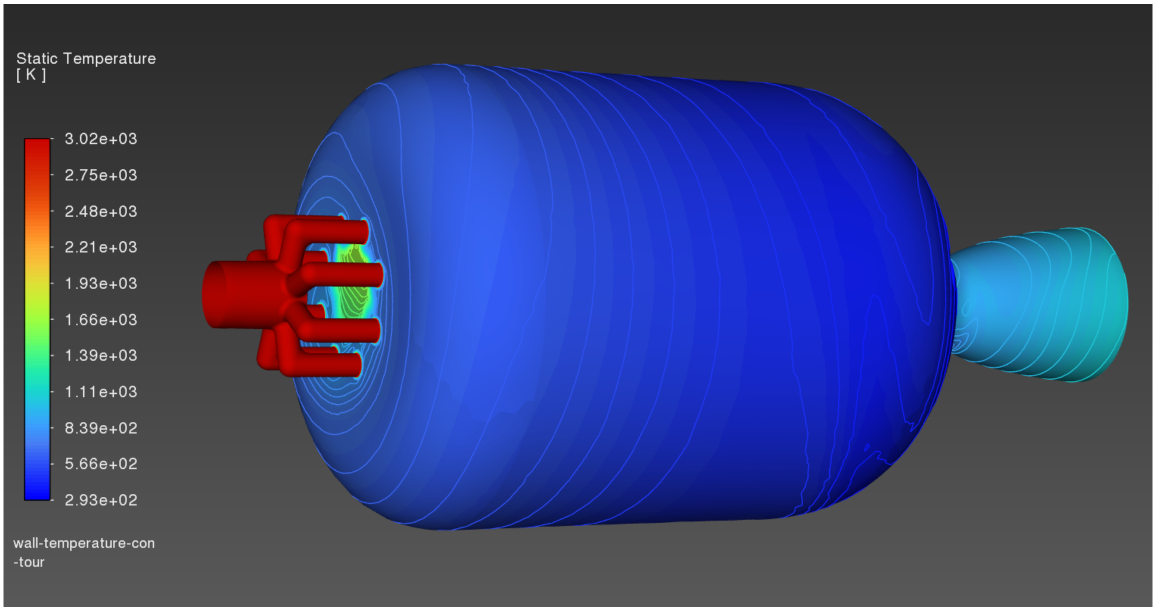


Figure 11: Static temperature distribution on the chamber and nozzle walls. Despite a 3000 K core, the chamber wall remains close to ambient over almost its entire area, with elevated temperature confined to the immediate vicinity of the heated fuel manifold where the hot stream physically enters. This is the central result of the study and is the direct visual confirmation of the vortex insulation effect.

4 Conclusion

A steady, compressible, three-dimensional cold-flow CFD study of the vortex-cooled rocket engine was carried out in Ansys Fluent. The results show that the bidirectional vortex forms as intended, with strong organized swirl filling the chamber and the expected low-velocity core. The converging-diverging nozzle chokes and accelerates the flow toward its design exit Mach number, and the residual swirl produces a non-uniform exit profile, which is the expected physical behavior. A two-species formulation captured how the axial and tangential streams mix through the chamber and at the outlet. With a representative hot core imposed, the chamber wall stayed well below the core temperature, which supports the main claim behind the vortex-cooled design that the cold outer vortex insulates the wall. The grid independence study showed that the level 3 mesh gives results within 1% of the finest mesh at a fraction of the cell count, and every reported solution uses that grid.

A few limitations should be stated. The turbulence closure is an eddy-viscosity model that assumes an isotropic turbulent viscosity, which is a poor assumption for strongly swirling flow, so the predicted swirl and insulation may be biased. Combustion is not modeled, so the hot core is imposed rather than generated, and the results should be read as a prediction conditioned on that imposed contrast. The working fluid density used the incompressible-ideal-gas law. This is accurate in the chamber, where the pressure is nearly uniform and the main results lie, but it does not capture the choked, supersonic nozzle flow correctly. A fully compressible solution using the full ideal-gas law would not converge in the time available, so the nozzle results are qualitative and no exit Mach number is claimed. The plenum is artificial and oversized, which produces recirculation at the outlet that is numerical rather than physical. The study also lacks experimental validation.

The most useful next step is to re-converge the case under the full ideal-gas law using a staged, limiter-assisted restart, which would make the nozzle results quantitative. Beyond that, a comparison against the oil-seeded flow visualization from the cold-flow test, a Reynolds stress turbulence model, and a more compact plenum would each strengthen the study.

Acknowledgments

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References

[1] T.-H. Shih, W. W. Liou, A. Shabbir, Z. Yang, and J. Zhu, "A New k-ε Eddy Viscosity Model for High Reynolds Number Turbulent Flows," *Computers & Fluids*, vol. 24, no. 3, pp. 227–238, 1995.

[2] P. J. Roache, "Perspective: A Method for Uniform Reporting of Grid Refinement Studies," *Journal of Fluids Engineering*, vol. 116, no. 3, pp. 405–413, 1994.

[3] American Society of Mechanical Engineers, *Standard for Verification and Validation in Computational Fluid Dynamics and Heat Transfer*, ASME V&V 20-2009, New York, 2009.

[4] W. Sutherland, "The Viscosity of Gases and Molecular Force," *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, vol. 36, no. 223, pp. 507–531, 1893.

[5] ANSYS, Inc., *ANSYS Fluent Theory Guide*, Release 2025 R2, ANSYS, Inc., Canonsburg, PA, 2025.